Explicit Gram certificates near a stationary Bell contact

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Abstract

An exact rational certificate places the previously separated C1 support inside the canonical cone used by the order-13744 candidate. A rational selector extends this certificate to an open five-parameter contact family: its compressed Grams have margin 2^{-22} at the reference source and remain positive for target variation below $1/3284140032$. A displayed one-parameter subfamily is certified on $|\sigma - 140| \le 2^{-42}$. The same selector fails at other regular sources, so universal adequacy remains unresolved.

The canonical C1 cone used by the order-13744 outer approximation contains the physical support that separates the two smaller candidates. More generally, an explicit rational selector proves membership for an open five-parameter family of stationary sources around that support. The selector does not work on the entire regular contact domain. Neither a universal converse for order 13744 nor a converse for C160 is proved here.

1 The canonical polynomial and its finite Gram representation

Write $c = (d, m, C, A, D, \alpha, \beta)$ and $\psi = (W, X, Y, Z)^T$. Set $D_u = 1 + u^2$, $D_v = 1 + v^2$, and define the entire scalar parent by

$$\begin{aligned} F_c = {} & 4m(vW - X + uvY - uZ)^2 + 4CD_v(uW - Y)^2 + 4AD_uD_vY^2 \\ & + (4D + 2d)(uW + uvX - Y - vZ)^2 + (4\alpha + 2d)D_u(vY - Z)^2 \\ & + (4\beta + 2d)D_v(X + uZ)^2 - 2dD_uD_vZ^2. \end{aligned} \tag{1}$$

The symbol $\mathcal{P}_c$ denotes the symmetric matrix such that $F_c = \psi^T \mathcal{P}_c \psi$. All source coordinates are fixed before the independent testing variables u, v, ψ are introduced.

For each of the four weights

$$(\alpha_g, \beta_g) = (0,0), (0,1), (1,0), (1,1), \qquad g = 0, 1, 2, 3,$$

start with the ordered features $u^r v^s \psi_i$, ordered lexicographically by $(i, r, s) \in \{0,1,2,3\} \times \{0,1\}^2$. Remove the following features:

g	removed features	n_g
0	$W,\ uX$	14
1	W	15
2	$W,\ uX$	14
3	uX	15

Let V_g be the retained feature vector. Define K_1 by $d \ge 0$ and the existence of $G_g \succeq 0$ satisfying the polynomial identity

$$(1 + u)(1 + v)F_c = \sum_{g=0}^{3} u^{\alpha_g} v^{\beta_g} V_g^T G_g V_g. \tag{2}$$

There are 450 upper-triangular entries and 148 coefficient equations. Each Gram entry contributes to one scalar monomial, so the coefficient equations have disjoint supports. Solving for one entry in each support leaves 302 free entries. These are the actual auxiliary coordinates in `C1_ENLARGED_PRIMAL.coo.json`. The four Gram blocks and $[d]$ have total order 59. The global compiler additionally imposes the six event signs and retains both canonical C1 orientations.

Equation (2) proves $F_c \geq 0$ for every $u, v \geq 0$ and every state vector. Its multiplier is strictly positive there. The exact sign gauges $(u, Y, Z) \mapsto (-u, -Y, -Z)$ and $(v, X, Z) \mapsto (-v, -X, -Z)$ extend this conclusion to signed finite angles. Continuity of the normalized physical operator gives the projective endpoints. Thus membership in K_1 is an all-state certificate, rather than a test restricted to its source contact.

2 An exact certificate for the separated support

The source is

$$\begin{aligned} c_* = \Bigg(1, \frac{1166523000}{31661820415017653}, \frac{322803000}{186199784281}, \frac{218288946200}{1359122513}, \\ \frac{90943434293}{222507771414}, \frac{248802430236}{3300726103}, \frac{205476973}{51646655494}\Bigg), \\ \theta_* = (1/13, 16, 140, -1/32, 1/50). \end{aligned} \tag{3}$$

Its homogeneous contact vector is $(140, -1/32, 1/50, 1)$, with last coordinate fixed to one. It is not a unit vector; dividing by its Euclidean norm gives the corresponding pure quantum state. The data file `C1_REPAIRED_TWO_FRAME_COUNTEREXAMPLE.json` supplies all 302 rational auxiliary entries y_*, all four actual Gram matrices, and rational factorizations

$$G_g(c_*, y_*) = F_g \Lambda_g F_g^T, \qquad \Lambda_g \succ 0, \qquad \operatorname{rank} F_g = n_g - 1. \tag{4}$$

The first $n_g - 1$ rows of each F_g are lower triangular with nonzero diagonal. Substitution into the current source pencil gives these exact matrices. Their quadratic forms give (2), and the four state kernel equations and both fixed-state angular derivatives vanish at (3). These are rational identities and positive rational diagonal entries; no optimization status is used.

Consequently $c_* \in K_1$. The separator for the old two-frame cone does not separate this enlarged cone.

For clarity, its support inequality in the eight observable coordinates is $\ell_{c_*} \geq 0$, where for a general c

$$\begin{aligned} \ell_c = d\left[1 - \frac{E_{00} - E_{01} + E_{10} + E_{11}}{2}\right] + m(1 + a_1 - b_1 - E_{11}) \\ + C(1 - a_1 + b_0 - E_{10}) + A(1 - a_0 + b_0 - E_{00}) \\ + D(1 - a_1 + b_1 - E_{11}) + \alpha(1 - a_0 - b_1 + E_{01}) \\ + \beta(1 + a_1 - b_0 - E_{10}). \end{aligned} \tag{5}$$

The exact nine affine coefficients are included in the selector data.

Here is the full physical restoration of this inequality. On a real two-qubit block choose

$$A_0 = B_0 = \sigma_z, \qquad A_1 = \frac{(1 - u^2)\sigma_z + 2u\sigma_x}{1 + u^2}, \qquad B_1 = \frac{(1 - v^2)\sigma_z + 2v\sigma_x}{1 + v^2}.$$

Replace the observable coordinates in (5) by the corresponding operators, denoting the result by $\ell_c(A, B)$. In the computational basis $\psi = (W, X, Y, Z)^T$, direct substitution gives

$$(1 + u^2)(1 + v^2)\psi^T \ell_c(A, B)\psi = F_c(u, v; \psi). \tag{6}$$

For example, the six positive-event amplitudes are proportional to $vW - X + uvY - uZ$, $uW - Y$, Y, $uW + uvX - Y - vZ$, $vY - Z$, and $X + uZ$, respectively. Thus (2), with its nonnegative

weights on the explicit testing domain $u, v \geq 0$, proves the operator inequality on every such block. The sign gauges and continuity cover all angles and endpoints. Real operator positivity implies complex operator positivity by separating the real and imaginary parts of a state vector.

Finally, two projective binary observables on one party decompose into invariant subspaces of dimension at most two: if $U = A_0 A_1$, then $A_0 U A_0 = U^*$, so the spectral subspaces of U pair at conjugate eigenvalues. On each resulting subspace the displayed Pauli form applies, including the one-dimensional endpoints. Applying this on both parties proves $\ell_c \geq 0$ for all quantum behaviors. Binary POVMs are covered by a common local projective dilation. Thus the source certificates here restore full-Q222 Bell inequalities. The positive-coordinate requirements specify the native source chart; validity follows from the restored full-parent identity, rather than from those coordinate signs alone.

There is also a full-dimensional cone with a linear auxiliary selector. Let e_i, $i = 1, \ldots, 6$, be the unit event rays and y_i their exact new-frame auxiliary vectors. For

$$\mathcal{W} = \{c : d \geq 0, \quad c_i - d(c_*)_i \geq 0, \ i = 1, \ldots, 6\}, \tag{7}$$

set

$$y(c) = d y_* + \sum_{i=1}^{6} (c_i - d(c_*)_i) y_i. \tag{8}$$

The corresponding matrices are the same nonnegative combination of the seven verified PSD matrices, proving $\mathcal{W} \subseteq K_1$. The exact 302×7 selector is supplied in `C1_REPAIRED_CHAMBER_AUXILIARY_MAP.json`.

3 A rational selector on an open contact domain

Put $\theta = (p, q, a, b, t)$, $\phi = (a, b, t, 1)^T$, and require $p, q > 0$ and $abt \neq 0$. Define the 6×7 polynomial contact matrix $\mathcal{J}(\theta)$ as follows: its first four rows give $\mathcal{P}_c(p, q)\phi/4$, and its last two rows give $\phi^T \mathcal{P}_{c,u}(p, q)\phi/4$ and $\phi^T \mathcal{P}_{c,v}(p, q)\phi/4$. These expressions are linear in c. Writing $\mathcal{J} = (\mathcal{J}_d \mid \mathcal{J}_e)$, set

$$c(\theta) = \big(1, -\mathcal{J}_e(\theta)^{-1} \mathcal{J}_d(\theta)\big), \qquad \Omega = \{p, q > 0, abt \neq 0, \det \mathcal{J}_e \neq 0, c_1, \ldots, c_6 > 0\}. \tag{9}$$

This is a rational definition from the entire parent. The data print all 42 entries of $\mathcal{J}$; each was independently rederived from (1). At the source in (3),

$$\det \mathcal{J}_e = -\frac{7033805705872535849475830606568681}{344646229622500000000} \neq 0. \tag{10}$$

Make the source-only change of variables

$$u = pU, \qquad v = qV, \qquad \psi = \operatorname{diag}(\phi)\chi. \tag{11}$$

In sector g, let $D_g = \operatorname{diag}(p^r q^s \phi_i)$ on the retained features. The normalized Gram is $\widetilde{G}_g = p^{\alpha_g} q^{\beta_g} D_g G_g D_g$. Its contact feature vector is $\mathbf{1}$. For $S_g = [I_{n_g - 1} \mid -\mathbf{1}]$, the matrices with this kernel are exactly

$$\widetilde{G}_g = S_g^T H_g S_g, \qquad H_g \in \mathbb{S}^{n_g - 1}, \qquad H_g \succeq 0 \iff \widetilde{G}_g \succeq 0. \tag{12}$$

Here H_g is the leading principal submatrix of $\widetilde{G}_g$. The equivalence follows also directly from the surjectivity of S_g.

Let $h \in \mathbb{R}^{392}$ list the upper entries of the four H_g, in lexicographic order (g, i, j). Let $\mathfrak{M}$ be the 148 scalar monomials $U^r V^s \chi_i \chi_j$ present in (2), in lexicographic order (r, s, i, j), $i \leq j$. Denote by E the 148×7 source coefficient matrix of $(1 + u)(1 + v)F_c$. The normalized target is

$$b(\theta)_{r,s,i,j} = p^r q^s \phi_i \phi_j (E c(\theta))_{r,s,i,j}. \tag{13}$$

The difference features $(V_{g,i} - V_{g,n_g})$ define a constant rational coefficient matrix $R \in \mathbb{Q}^{148\times 392}$ such that the entire matching equation is $Rh = b(\theta)$.

Here are the exact linear identities needed below. Let C have columns

$$C_{:,r,s,i,j} = (\delta_{0i} + \delta_{0j}, \ldots, \delta_{3i} + \delta_{3j}, r, s)^T. \tag{14}$$

Then

$$CR = 0, \quad \operatorname{rank} C = 6, \quad \operatorname{rank} R = 142, \quad \operatorname{im} R = \ker C. \tag{15}$$

The first identity follows because a product of two difference features has zero value and first derivatives at the contact. For an explicit certificate of the rank assertion, choose the lexicographically first 142 independent rows I of R and the first 142 independent columns J of R_I. The supplied rational matrix

$$B = R_{I,J}, \qquad P = B^{-1} \tag{16}$$

obeys $BP = I_{142}$. Its 1387 nonzero entries and the complete coefficient maps are in `C1_ENLARGED_REGULAR_CONTACT_SYSTEM.json`. Thus (15) is certified by explicit finite matrix identities. Since the complete contact equations vanish,

$$Cb(\theta) = 0 \qquad (\theta \in \Omega). \tag{17}$$

The multiplier derivatives contribute zero because the parent value is zero at the contact. In particular no state equation has been omitted.

Let h_* be obtained from (4) by (11)–(12), let $b_* = b(\theta_*)$, and let J^c be the remaining 250 columns. Define the rational selector

$$h_{J^c}(\theta) = (h_*)_{J^c}, \qquad h_J(\theta) = (h_*)_J + P\big(b(\theta)_I - (b_*)_I\big). \tag{18}$$

Equations (15)–(17) imply $Rh = b(\theta)$: the difference has zero I rows and lies in the 142-dimensional space $\operatorname{im} R$, on which those rows are injective.

Theorem 1 (Explicit open domain)**.** *Let*

$$\mathcal{U} = \left\{\theta \in \Omega : \|b(\theta) - b_*\|_\infty < \frac{1}{3284140032}\right\}. \tag{19}$$

For every $\theta \in \mathcal{U}$, (18) *gives four positive definite compressed Grams and hence an exact certificate $c(\theta) \in K_1$. The set $\mathcal{U}$ contains an open neighborhood of θ_*, and its image contains a five-dimensional family of distinct stationary sources.*

Proof. The first $n_g - 1$ rows of the factors in (4) are invertible. The source normalizations are invertible too, so every $(H_g)_*$ is positive definite. Exact rational inverse traces give the bounds

g	0	1	2	3
$\operatorname{tr}((H_g)_*^{-1}) \le$	2^{12}	2^{17}	2^{22}	2^{17}
L_g	364	437/2	409/2	783/2

The exact inverse traces are supplied with the selector data. Since $\lambda_{\min}(H) \ge 1/\operatorname{tr}(H^{-1})$ for $H \succ 0$,

$$(H_g)_* \succeq 2^{-22} I. \tag{20}$$

For completeness, the displayed row bounds use the following explicit definition. For a pivot entry indexed by (g, i, j), put $a_{gij} = \sum_{k=1}^{142} |P_{\mathrm{pos}(g,i,j),k}|$; put $a_{gij} = 0$ for a fixed free entry and reflect it symmetrically. Then $L_g = \max_i \sum_j a_{gij}$. Thus (18) gives

$$\|H_g(\theta) - (H_g)_*\|_2 \le \|H_g(\theta) - (H_g)_*\|_\infty \le \frac{783}{2}\|b(\theta) - b_*\|_\infty < 2^{-23}.$$

Together with (20), this proves $H_g(\theta) \succ 2^{-23} I$.

Restore the original Grams by the explicit rational formula

$$G_g(\theta) = p^{-\alpha_g} q^{-\beta_g} D_g^{-1} S_g^T H_g(\theta) S_g D_g^{-1}. \tag{21}$$

All denominators are nonzero on Ω. They depend only on the source; the restored matrices satisfy the formal polynomial identity (2) in independent u, v, ψ. The original auxiliary $y_j(\theta)$ is the entry of $G_g(\theta)$ at the jth listed free position of the literal source pencil.

The inequalities defining Ω are strict at (3) and the target is continuous there, so $\mathcal{U}$ contains an open neighborhood. Differentiating the six contact equations gives

$$\partial_{\theta_k} c_e = -\mathcal{J}_e^{-1} (\partial_{\theta_k} \mathcal{J}) c.$$

At the seed this 6×5 rational matrix has rank five. This proves the asserted local dimension without identifying testing variables with source variables. □

4 A short rational source family and the selector's limitation

Fix the four contact coordinates $(p, q, b, t) = (1/13, 16, -1/32, 1/50)$ and vary $a = \sigma$. Put $Q_\sigma = 83200\sigma^2 + 17476053\sigma + 119$. Solving the entire contact system gives $d = 1$ and

$$\begin{aligned}
m(\sigma) &= \frac{3499569000}{(166400\sigma - 219) Q_\sigma}, \\
C(\sigma) &= \frac{177021000(2\sigma - 1)}{(50\sigma - 13) Q_\sigma}, \\
A(\sigma) &= \frac{2600(12800\sigma^2 + 7122\sigma - 5219)}{Q_\sigma}, \\
D(\sigma) &= -\frac{2080000\sigma^3 + 2680525\sigma^2 - 218723632\sigma - 621061}{2(25\sigma - 5219) Q_\sigma}, \\
\alpha(\sigma) &= \frac{3204(83200\sigma^2 + 53\sigma + 119)}{17 Q_\sigma}, \\
\beta(\sigma) &= -\frac{13(121600\sigma^2 - 17362101\sigma - 83623)}{38 Q_\sigma}.
\end{aligned} \tag{22}$$

These formulas are exact stationary-source identities, including all four state equations and both angular equations. Their contact minor is

$$\det \mathcal{J}_e(\sigma) = \frac{6165476003(25\sigma - 5219)(50\sigma - 13)(166400\sigma - 219) Q_\sigma}{344646229622500000000}, \tag{23}$$

which is nonzero throughout $139 \le \sigma \le 141$.

All six coefficients are positive for $139 \le \sigma \le 141$. Here is a direct uniform sign verification. Write each displayed fraction as $N_i(s)/D_i(s)$ after $\sigma = 140 + s$ and expand the numerator and denominator, without cancelling their displayed integer factors. Each constant has greater absolute value than the sum of the remaining coefficient magnitudes. The positive differences are

i	$\lvert N_i(0)\rvert - \sum_{k\ge1} \lvert N_{ik}\rvert$	$\lvert D_i(0)\rvert - \sum_{k\ge1} \lvert D_{ik}\rvert$
m	3499569000	93348433945876566
C	49034817000	27997200202682
A	645496641400	4036512286
D	5605722176566	13952538819296
α	5149943908344	68620708862
β	397933432	153387466868

The constant numerator and denominator have the same sign in every row. This proves the assertion on $|s| \le 1$.

For an explicit interval on which the selector is guaranteed to work, write each target entry in (13) as a reduced fraction $N(s)/D(s)$. Let $\|N\|_1 = \sum |N_k|$ and $\|N'\|_1 = \sum k|N_k|$, and set $d_0 = |D_0| - \sum_{k\ge1} |D_k| > 0$. For $|s| \le 1$,

$$\left| \frac{\mathrm{d}}{\mathrm{d}s} \frac{N(s)}{D(s)} \right| \le \frac{\|N'\|_1 \|D\|_1 + \|N\|_1 \|D'\|_1}{d_0^2}.$$

Taking the maximum over the explicit target entries gives the exact bound

$$B = \frac{1867461120440940797743314026982967720800 0}{3739970341355847909429761599855495261 3} < 500.$$

Therefore every rational or real σ satisfying

$$|\sigma - 140| \le 2^{-42} \tag{24}$$

has $\|b(\sigma) - b_*\|_\infty \le B2^{-42} < 1/(2 \cdot 3284140032)$ and is certified by (18)–(21).

The same fixed choice of 250 free entries cannot be asserted on the whole regular contact domain. For example, (22) at $\sigma = 70$ has six strictly positive coefficients and a nonsingular contact matrix. The restored compressed selector has

$$(H_2(70))_{00} = -\frac{427612930275749907469024085268927781247479080275190191 3517}{945649597692176417975102921534884031878798850000000000 0} < -400. \tag{25}$$

It is thus not PSD. Even at $\sigma = 141$, the first nonpositive Schur pivot in H_0 is its sixth pivot and is strictly negative; the exact rational value is supplied. These calculations refute the universal use of *this selector*. They do not refute membership in K_1: other values of the 250 free coordinates remain available.

5 The precise remaining adequacy question

The general C1 contact problem has an explicit finite form. For an arbitrary regular contact, let $h_{J^c} = z \in \mathbb{R}^{250}$ and set

$$h_J = P\big(b_I - R_{I,J^c} z\big). \tag{26}$$

Its four matrices have orders $(13, 14, 13, 14)$. The other canonical orientation is the rational contact map

$$\tau(p, q, a, b, t) = \left(p^{-1}, q, \frac{qa - b}{t + q}, \frac{-a - qb}{t + q}, \frac{1 - qt}{t + q} \right), \tag{27}$$

with source permutation $\sigma c = (d, C, m, \alpha, \beta, A, D)$. The exact parent congruence behind this map is

$$u^2(1 + v^2)\mathcal{P}_{\sigma c}(1/u, v) = G(v)\mathcal{P}_c(u, v)G(v)^T, \quad G(v) = \begin{pmatrix} v & -1 & 0 & 0 \\ -1 & -v & 0 & 0 \\ 0 & 0 & -v & 1 \\ 0 & 0 & 1 & v \end{pmatrix}.$$

On the regular two-orientation domain, the universal statement still needed is that for every contact there is an orientation and a value of z for which all four matrices in (26) are PSD. The construction above proves this statement on $\mathcal{U}$ with one explicit value of z. It supplies no uniform selector on the complement. Completing C1 would also leave the independent whole-C2 converse needed by the order-13744 compiler.

In particular, a positive Gram margin at one contact establishes an open family, but it neither gives a finite cover of escaping or singular contacts nor licenses a constant free-coordinate choice everywhere. Equation (25) identifies this limitation exactly. No conclusion about C160 reverse inclusion is inferred from the C1 certificate.